National University of Computer and Emerging Sciences

Islamabad Campus

Probability & Statistics Sessional-2 Solution

Q1: **[16 marks]**

(a) Does $E(XY) = E(X).E(Y)$ always hold? If not, when does it hold ? (1)

Solution $E(XY) = E(X).E(Y)$ only if X and Y are independent random variables.

(b) A random committee of size 3 is selected from 4 doctors and 5 nurses. If *X* represents the number of doctors on the committee.

(i) Find the probability distribution of X. (3)

(ii) Find $E(X)$ *and* $V(X)$. (3)

Solution Let X denote number of doctors selected in the committee.

Here $Number\ of\ doctors = 4,\ Number of\ Nurses = 5,\ Selected\ members = 3$

X	P(X)	X P(X)	X2 P(X)
0	$\frac{{}^4C_0 \times {}^5C_3}{{}^9C_3} = \frac{10}{84} = \frac{5}{42}$	0	0
1	$\frac{{}^4C_1 \times {}^5C_2}{{}^9C_3} = \frac{40}{84} = \frac{10}{21}$	$\frac{40}{84}$	$\frac{40}{84}$
2	$\frac{{}^4C_2 \times {}^5C_1}{{}^9C_3} = \frac{30}{84} = \frac{5}{14}$	$\frac{60}{84}$	$\frac{120}{84}$
3	$\frac{{}^4C_3 \times {}^5C_0}{{}^9C_3} = \frac{4}{84} = \frac{1}{21}$	$\frac{12}{84}$	$\frac{36}{84}$
SUM	1	112/84	196/84

$Now\ E(X) = \sum X\ P(X) = 112/84 = 1.33 \quad also\ E(X^2) = \sum X^2 P(X) = 196/84 = 2.33$

$V(X) = E(X^2) - (E(X))^2 = 2.33 - (1.33)^2 = 0.5644$

(c) The probability density function of a random variable *X* is given by

$$f(x) = \begin{cases} ax + bx^2 & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

(*i*) *If* $E(X) = 1/9$, *find the values of a and b.* (3)

(*ii*) *Find cumulative distribution function.* (3)

(*iii*) *Find* $P(0.3 \le X \le 0.8)$ *using pdf and cdf*. (3)

Solution (i) values of a and b.

$As\ we\ know\ that\ f(x)\ is\ a\ proper\ pdf \Rightarrow \int_0^1 (ax + bx^2)dx = 1 \Rightarrow 3a + 2b = 6 \text{ -----(1)}$

$Also\ it\ is\ given\ that\ E(X) = 1/9. \Rightarrow \int_0^1 x(ax + bx^2)dx = 1/9 \Rightarrow 4a + 3b = 12/9 \text{ ----(2)}$

Solving (1) and (2) we get $a = 46/3 = 15.33 \quad and \quad b = -20$

(ii) Cumulative distribution function

$$cdf\ is\ given\ by\ F(x)=\int_0^x\left(\frac{46}{3}x-20x^2\right)dx=\frac{23}{3}x^2-\frac{20}{3}x^3$$

$$So\quad F(x)=\begin{cases}0 & if\ \ x<0\\ \frac{23}{3}x^2-\frac{20}{3}x^3 & if\ \ 0\le x\le 1\\ 1 & if\ \ x>1\end{cases}$$

(iii) Finding Probability using pdf and cdf

$$U\sin g\ \ pdf\ P(0.3\le X\le 0.8)=\int_{0.3}^{0.8}\left(\frac{46}{3}x-20x^2\right)dx=\left|\frac{23}{3}x^2-\frac{20}{3}x^3\right|_{0.3}^{0.8}=\frac{59}{60}=0.9833$$

$$U\sin g\ \ cdf\ \ P(0.3\le X\le 0.8)=F(0.8)-F(0.3)$$

$$=\left(\frac{23}{3}(0.8)^2-\frac{20}{3}(0.8)^3\right)-\left(\frac{23}{3}(0.3)^2-\frac{20}{3}(0.3)^3\right)=\frac{59}{60}=0.9833$$

Q2: **[18 marks]**

(a) Let X denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

		x		
$f(x,y)$		1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

I. Evaluate the marginal distributions of X and Y. (3)

II. Find $E(X)$, $E(Y)$ *and* $E(XY)$ (3)

III. Find $COV(X,Y)$ and check whether X and Y are independent or not. (3)

Solution

(a) *Marginal Distributions*

$g(x)$: 0.1 0.35 0.55

$h(y)$: 0.2 0.5 0.3

(b) $E(X)=\sum X\ g(X)=2.45$

$E(Y)=\sum Y\ h(Y)=3.2$

$E(XY)=\sum\sum XY\ f(x,y)=7.85$

$(c)\ Cov(X,Y)=E(XY)-E(X).E(Y)=7.85-2.45\times 3.2=0.01$

$Since\ \ Cov(X,Y)\neq 0\Rightarrow X\ \ and\ \ Y\ \ are\ not\ independent.$

(b) Consider the following joint probability density function of the random variables X and Y :

$$f(x,y)=\begin{cases}\dfrac{3x-y}{9} & 1<x<3,\ 1<y<2\\ 0 & elsewhere\end{cases}$$

I. Find marginal probability functions of X and Y. (6)

II. Find $P(X+Y<3)$ (3)

Solution

(i) M arg$inal$ Pr$obability\ Functions$

$$g(x)=\int_1^2 \frac{3x-y}{9}\,dy=\frac{2x-1}{6}$$

$$h(y)=\int_1^3 \frac{3x-y}{9}\,dx=\frac{12-2y}{9}$$

(ii) $P(X+Y<3)$

$$P=\int_1^2\int_1^{3-y}\frac{3x-y}{9}\,dx\,dy=\frac{4}{27}=0.1481\quad (Moving\ Horizontally)$$

$$P=\int_1^2\int_1^{3-x}\frac{3x-y}{9}\,dy\,dx=\frac{4}{27}=0.1481\quad (Moving Vertically)$$

Q3: **[22 marks]**

(a) Let $X\sim Poisson\,(\lambda)$, Show that the MDF of the Poisson distribution is : $M_X(t)=e^{\lambda(e^t-1)}$

Using the MGF, find $E(X)$ *and* $V(X)$. (4+3+3)

Solution *MGF OF POISSON DISTRIBUTION*

Moment Generating Function is given by

$$M_X(t)=E(e^{tx})=\sum_{x=0}^{\infty}e^{tx}P(X=x)=\sum_{x=0}^{\infty}\frac{e^{-\lambda}e^{tx}\lambda^x}{x!}=e^{-\lambda}\sum_{x=0}^{\infty}\frac{(e^t\lambda)^x}{x!}$$

$$=e^{-\lambda}\left[\frac{(e^t\lambda)^0}{0!}+\frac{(e^t\lambda)^1}{1!}+\frac{(e^t\lambda)^2}{2!}+\ldots\right]=e^{-\lambda}\left[1+\frac{(e^t\lambda)^1}{1!}+\frac{(e^t\lambda)^2}{2!}+\ldots\right]$$

$$=e^{-\lambda}.e^{e^t\lambda}=e^{-\lambda+e^t\lambda}=e^{\lambda(e^t-1)}$$

Mean usin$g$ Moment Generating Function is given by

$$E(X)=\left[\frac{d}{dt}M_X(t)\right]_{t=0}=\left[\frac{d}{dt}(e^{\lambda(e^t-1)})^n\right]_{t=0}=\left[\lambda e^t e^{\lambda(e^t-1)}\right]_{t=0}=\lambda$$

Variance usin$g$ Moment Generating Function. For var*iance we shall first find* $E(X^2)$

$$E(X^2)=\left[\frac{d^2}{dt^2}M_X(t)\right]_{t=0}=\left[\frac{d}{dt}\left(\lambda e^t e^{\lambda(e^t-1)}\right)\right]_{t=0}=\lambda\left[e^t e^{\lambda(e^t-1)}+\lambda e^t e^{\lambda(e^t-1)}.e^t\right]_{t=0}=\lambda[1+\lambda]=\lambda+\lambda^2$$

Now $V(X)=E(X^2)-(E(X))^2=\lambda+\lambda^2-(\lambda)^2=\lambda$

(b) A cloud server processes user requests. Each request is successful with probability $p = 0.6$ and failures occur independently

(i) In a batch of 10 independent requests, find the probability that exactly 7 requests are successful. (2)

Solution: *Here* $X \sim$ *Binomial Distrbution also* $n = 10,\ p = 0.6,\ q = 1 - p = 1 - 0.6 = 0.4$

Now $P(X = 7) = {}^{10}C_7 (0.6)^7 (0.4)^3 = 0.2150$

(ii) Find the probability that the first successful request occurs on the 4th attempt. (2)

Solution: *Here* $X \sim$ *Geometric Distrbution also* $p = 0.6,\ q = 1 - p = 1 - 0.6 = 0.4$

Now $P(X = 4) = p\,q^{x-1} = (0.6)(0.4)^{4-1} = 0.0384$

(iii) Find the probability that the 3rd successful request occurs on the 6th attempt. (2)

Solution: *Here* $X \sim$ *Negative Binomial Distrbution also* $p = 0.6,\ q = 1 - p = 0.4,\ r = 3$

Now $P(X = 6) = {}^{x-1}C_{r-1} p^r q^{x-r} = {}^{6-1}C_{3-1} (0.6)^3 (0.4)^{6-3} = {}^5C_2 (0.6)^3 (0.4)^3 = 0.13824$

(iv) A fixed batch of 20 requests contains 12 successful ones. If 5 requests are selected randomly without replacement, find the probability that exactly 3 are successful. (2)

Solution: *Here we use combinations*

also total requests $= 20,$ *Successful* $= 12,$ *Unsuccessful* $= 8,$ *Select* $= 5$

Now $P(X = 3) = \dfrac{{}^{12}C_3 . {}^8C_2}{{}^{20}C_5} = \dfrac{6160}{15504} = \dfrac{385}{969} = 0.3973$

(v) Requests arrive at an average rate of 4 per minute. Find the probability that exactly 6 requests arrive in one minute. (2)

Solution: *Here* $X \sim$ *Poisson Distrbution also* $\lambda = 4$

Now $P(X = 6) = \dfrac{e^{-4} 4^6}{6!} = 0.1042$

(vi) Each request can result in one of three outcomes: Success (S), Delay (D), or Failure(F), with probabilities 0.5, 0.3, and 0.2. In 8 independent requests, find the probability of observing: 4 successes, 2 delays and, 2 failures. (2)

Solution: *Here* $X \sim$ *Multinomial Distrbution, Let us define the random variables first.*

Let X_1 *denote the event that there is Success*

Let X_2 *denote the event that there is Delay*

Let X_3 *denote the event that there is Failure*

Here $n = 8,\ X_1 = 4,\ X_2 = 2,\ X_3 = 2,\ p_1 = 0.5,\ p_2 = 0.3,\ p_3 = 0.2$

Then $P(X_1 = 4,\ X_2 = 2,\ X_3 = 2) = \dfrac{8!}{4! \times 2! \times 2!} (0.5)^4 (0.3)^2 (0.2)^2 = \dfrac{189}{2000} = 0.0945$

Q4: **[24 marks]**

(a) The MGF of Gamma Distribution is given by

$$M_X(t)=\left(\frac{\lambda}{\lambda-t}\right)^{\alpha}$$

Find CGF of Gamma Distribution, then find mean and variance using CGF. (3+3+3)

Solution ***CGF OF GAMMA DISTRIBUTION***

Cumulant Generating Function is given by

$$K_X(t)=\ln(M_X(t))$$

$$=\ln\left(\frac{\lambda}{\lambda-t}\right)^{\alpha}=\alpha[\ln\lambda-\ln(\lambda-t)]$$

Mean using Cumulant Generating Function is given by

$$E(X)=\left[\frac{d}{dt}K_X(t)\right]_{t=0}$$

$$=\left[\frac{d}{dt}\alpha(\ln\lambda-\ln(\lambda-t))\right]_{t=0}$$

$$=\alpha\left[0-\frac{1}{\lambda-t}\times-1\right]_{t=0}=\left[\frac{\alpha}{\lambda-t}\right]_{t=0}=\frac{\alpha}{\lambda}$$

Variance using Cumulant Generating Function is given by

$$V(X)=\left[\frac{d^2}{dt^2}K_X(t)\right]_{t=0}$$

$$=\left[\frac{d}{dt}\left(\frac{\alpha}{\lambda-t}\right)\right]_{t=0}$$

$$=\left[(-\alpha)(\lambda-t)^{-2}(-1)\right]_{t=0}$$

$$=\left[\alpha(\lambda-t)^{-2}\right]_{t=0}=\frac{\alpha}{\lambda^2}$$

(b) The masses of packages from a particular machine are normally distributed with a mean of 200 g and a standard deviation of 2 g. Find the probability that a randomly selected package from the machine weighs between 198.5 g and 199.5 g (3)

Solution Let X denote masses of Packages

Here $X\sim N(200,4)\Rightarrow\mu=200,\ \sigma=2$

$P(198.5<X<199.5)=?$

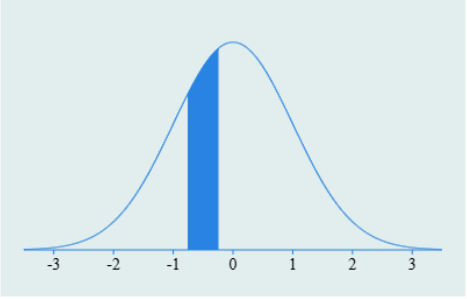
$$Z_1=\frac{X-\mu}{\sigma}=\frac{198.5-200}{2}=-0.75$$

$$Z_2=\frac{X-\mu}{\sigma}=\frac{199.5-200}{2}=-0.25$$

So $P(-0.75<Z<-0.25)$

$=\Phi(-0.25)-\Phi(-0.75)$

$=0.4013-0.2266=0.1747$



(c) A random variable X has a mean $\mu = 8$ and variance $\sigma^2 = 9$ and an unknown probability distribution. Using Chebyshev's Inequality, find.

$(i)\ P(-4 < X < 20)$ (3)

$(ii)\ P(|X - 8| < 6)$ (3)

Solution

Here $\mu = 8$ & $\sigma^2 = 9$. Chebyshev's Inequality is given by $P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$

$(a)\ P(-4 < X < 20) \Rightarrow \mu - k\sigma = -4$ *and* $\mu + k\sigma = 20$

subtracting two equations we get $2k\sigma = 24 \Rightarrow 2k = \frac{24}{\sigma} = \frac{24}{3} = 8 \Rightarrow k = 4$

So $P(-4 < X < 20) \geq 1 - \frac{1}{k^2} = 1 - \frac{1}{4^2} = 1 - \frac{1}{16} = \frac{15}{16}$ *So* $P(-4 < X < 20) \geq \frac{15}{16}$

$(b)\ P(|X - 8| < 6)$

$= P(-6 < X - 8 < 6)$

$= P(2 < X < 14)$

$\Rightarrow \mu - k\sigma = 2$ *and* $\mu + k\sigma = 14$

subtracting two equations we get $2k\sigma = 12 \Rightarrow 2k = \frac{12}{\sigma} = \frac{12}{3} = 4 \Rightarrow k = 2$

So $P(2 < X < 14) \geq 1 - \frac{1}{k^2} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4}$, *So* $P(2 < X < 14) \geq \frac{3}{4}$

(d) Let $X_1, X_2, \ldots, X_{10}$ be independent random variables, each uniformly distributed over $(0,1)$. Let $S = X_1 + X_2 + \ldots + X_{10}$ then using central limit theorem calculate $P(S > 6)$. (6)

Solution

Here $X \sim$ *Uniform Distribution with* $a = 0$ *and* $b = 1$.

Here mean $\mu = \frac{a+b}{2} = \frac{0+1}{2} = 0.5$ & $\sigma^2 = \frac{(b-a)^2}{12} = \frac{(1-0)^2}{12} = \frac{1}{12}$.

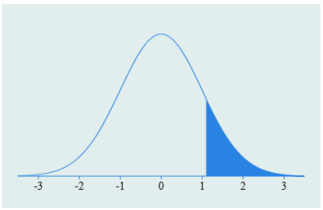
Now $S = X_1 + X_2 + \ldots + X_{10}$

Using Central Limit Theorem *If* $S = X_1 + X_2 + \ldots + X_n$ *then* $S \sim N(n\mu, n\sigma^2)$

Here $S \sim N((10)(0.5), (10)(1/12)) \Rightarrow S \sim N(5, 5/6)$

So mean of S is 5 and variance of S is 5/6.

Now $P(S > 6) = ?$ *where* $S \sim N(5, 5/6)$



Now $Z = \frac{6-5}{\sqrt{5/6}} = \frac{1}{\sqrt{5/6}} = 1.10$

Now $P(Z > 1.10) = 1 - \Phi(1.10) = 1 - 0.8643 = 0.1357$